

Fine-tuning a Small LLM with LoRA for Automated Paper Review Insights

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1 Introduction

Peer review remains a foundational pillar of the scientific process, ensuring rigor, validity, and quality through independent expert evaluation (Bornmann, 2011). However, as the number of submissions to journals and conferences continues to grow, the peer review system faces increasing strain, leading to delayed editorial decisions, inconsistent feedback, and reviewer fatigue (Aczel et al., 2025). In parallel, recent advancements in Large Language Models (LLMs) have opened new opportunities for assisting or partially automating aspects of the review process. Current approaches vary in cost, flexibility, and performance, and include: (1) prompt-based zero-shot generation, (2) full supervised fine-tuning on human review data, and (3) parameter-efficient fine-tuning such as Low-Rank Adaptation (LoRA). Prompt engineering involves using general-purpose LLMs (e.g., GPT-4) in a zero-shot setting to generate review-like text given paper abstracts and task-specific instructions (Gilardi et al., 2023). This approach is simple and requires no additional training, but its outputs are highly sensitive to prompt design and lack domain adaptation. In contrast, full supervised fine-tuning retrains all parameters of an LLM on large corpora of human-written reviews, producing highly specialized models (Zhou et al., 2022). While this method can achieve strong task-specific performance, it demands significant computational resources and large-scale annotated datasets. Parameter-efficient fine-tuning techniques, such as LoRA (Hu et al., 2021), offer a compromise by updating only a small number of low-rank parameters within the model, enabling task adaptation even with limited data and compute.

This project investigates the feasibility of using LoRA to fine-tune a small open-source LLM (e.g., LLaMA 3.2B) on real-world peer review data. Specifically, we aim to construct a lightweight model capable of generating structured outputs—highlighting the strengths and weaknesses of a scientific paper—while also producing numeric predictions (rating and confidence) aligned with human annotations. Our dataset was derived from the OpenReview platform, which provides open-access peer reviews linked to scientific submissions. The experimental design compares zero-shot predictions of the base model against LoRA fine-tuned outputs, evaluated through regression metrics (MAE, RMSE, R^2 , Pearson’s r) and semantic similarity measures (BERTScore). By contrasting these approaches, we assess whether parameter-efficient fine-tuning can improve alignment with human evaluators while remaining computationally affordable. Unlike studies that rely on large proprietary systems (Tyser et al., 2024), this work emphasizes accessibility, reproducibility, and efficiency, contributing to the emerging field of Automated Scholarly Paper Review (ASPR).

⁰Project repository on GitHub: https://github.com/camillabonomo02/ML_project.git

2 Methods

This project employed a multi-stage methodology designed to explore whether a lightweight large language model (Meta’s LLaMA 3.2B Instruct model) fine-tuned with Low-Rank Adaptation (LoRA) can generate structured peer reviews that approximate human assessment. The pipeline was structured to ensure reproducibility and to provide a rigorous framework for both data preparation and model evaluation.

1. Data collection and preprocessing

The dataset was derived from the OpenReview repository of conference submissions (`tp_2017conference.xlsx`). Each entry contained a paper title, abstract, reviewer comments, rating, confidence score, and final decision. The raw Excel file was first cleaned by removing rows with missing values in essential fields (title, abstract, review) and by standardizing text through regular expressions to eliminate non-printable characters, HTML tags, and formatting artifacts. Duplicate entries based on paper titles were consolidated by concatenating multiple reviews into a single entry. The dataset was then split into training (72%), validation (8%), and test (20%) sets using `train_test_split` from `scikit-learn`. For subsequent regression analysis, the original `rate` and `confidence` fields, which appeared as annotated strings (e.g., “Rating:###7: Good paper, accept”), were normalized into numeric columns (`rating_num`, `confidence_num`). The `decision` field was removed to prevent label leakage, ensuring that the model did not simply learn to justify acceptance or rejection outcomes.

2. Zero-Shot baseline

Before fine-tuning, a zero-shot baseline was established using an instruction-tuned LLM (LLaMA-3.2-3B-Instruct). To guarantee comparability, a rigid prompt template was employed requiring the model to generate outputs in a fixed structure:

Strengths:

1. ...
2. ...

Weaknesses:

1. ...
2. ...

Rating: <number>

Confidence: <number>

This format enforced consistency across generated reviews and ensured the simultaneous production of structured text and numeric predictions. The zero-shot model outputs served two purposes: (i) they acted as pseudo-labels to bootstrap fine-tuning on limited human-annotated data, and (ii) they provided a baseline for performance evaluation.

3. Fine-Tuning with LoRA

LoRA was employed to fine-tune the base model efficiently. By introducing low-rank adaptation matrices into attention layers, LoRA enables training with significantly fewer parameters while preserving the frozen base model weights. The fine-tuning dataset paired paper titles and abstracts with human **and** zero-shot generated reviews. Inputs were tokenized with a maximum length of 300 tokens, and labels were aligned with target review outputs. Training employed the Hugging Face **Trainer** API with 4-bit quantization for memory efficiency, a batch size of 2, and three epochs. The structured format of the

outputs allowed the model to learn not only to generate strengths and weaknesses but also to provide numerical ratings aligned with human annotations.

4. Evaluation

Model performance was evaluated on the held-out test set using two complementary approaches. First, regression metrics (MAE, RMSE, R^2 , and Pearson’s r) were computed by comparing predicted **rating** and **confidence** values against ground-truth human scores, providing a quantitative measure of alignment with human evaluations. Second, semantic similarity between the generated and human-written reviews was assessed using BERTScore. This evaluation was performed for both the zero-shot and fine-tuned models, enabling a direct comparison of how each model’s generated content aligned semantically with human reviews. By comparing both BERTScores, we can gauge whether fine-tuning improves the model’s ability to generate more coherent and relevant reviews, or if it merely reproduces patterns seen in the zero-shot outputs. However, while BERTScore offers valuable insights into semantic similarity, it is limited by its reliance on pre-trained language models, which may not fully capture domain-specific nuances or reviewer intent. Thus, BERTScore provides an important but supplementary evaluation that should be considered alongside traditional regression metrics.

5. Challenges and limitations

Several challenges were encountered during the process. The human review dataset was relatively small and noisy, with heterogeneous writing styles and variable annotation quality. While zero-shot pseudo-labeling helped expand the dataset, it introduced the risk of propagating biases and generic formulations present in the base LLM. Furthermore, extracting structured numeric predictions reliably required carefully engineered prompts, as unconstrained models often produced incomplete or inconsistent outputs. Finally, computational constraints necessitated the use of quantization and LoRA rather than full fine-tuning, which limited the exploration of larger architectures.

3 Results

This section presents the outcomes of the evaluation of both the zero-shot and fine-tuned models. The focus is on two main aspects: (1) predictive accuracy of numerical ratings and confidence scores, (2) semantic alignment of generated reviews with human-written reviews.

Quantitative Evaluation

Table 1 reports the regression-based performance of the models using human-provided ratings and confidence scores as ground truth. Metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Pearson correlation. The analysis was based on 72 usable zero-shot samples and 69 fine-tuned samples.

Task	MAE	RMSE	R^2	Pearson
Zero-shot Rating	1.26	1.58	-0.02	0.45
Fine-tuned Rating	1.28	1.54	-0.06	0.49
Zero-shot Confidence	0.83	1.08	-1.02	0.09
Fine-tuned Confidence	0.81	1.10	-0.94	0.15

Table 1: *Regression metrics for predicted ratings and confidence scores. Lower MAE/RMSE and higher correlation value indicate better alignment with human annotations.*

Fine-tuning produced marginal improvements in predictive consistency, particularly for the rating task, where RMSE decreased from 1.58 to 1.54 and Pearson’s correlation increased from 0.45 to 0.49. Although the negative R^2 values indicate limited explanatory power relative to the dataset mean,

these results remain consistent with the expected variability and subjectivity of peer-review scores. For confidence prediction, results remain weak across all metrics, confirming that textual cues alone are insufficient to infer reviewer certainty. The aggregated error comparison is visualized in Figure 1. The bar chart highlights that fine-tuning reduces variance in rating prediction (slightly lower

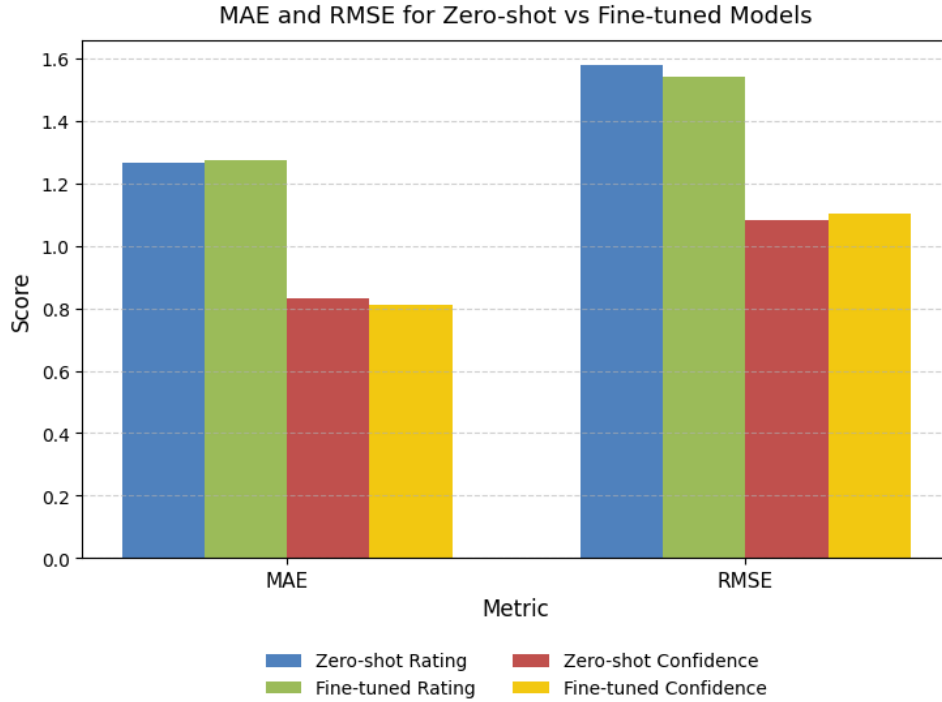


Figure 1: Comparison of MAE and RMSE for zero-shot and fine-tuned models across rating and confidence prediction tasks.

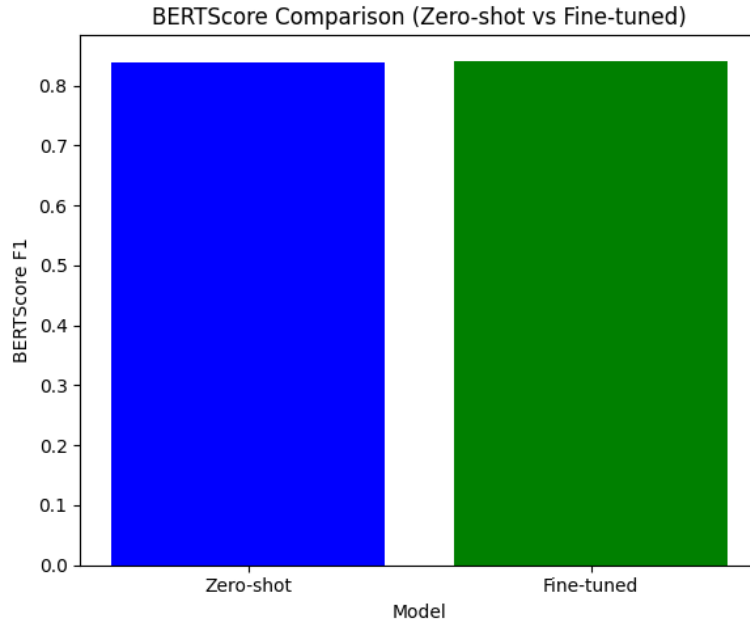


Figure 2: BERTScore F1 comparison between zero-shot and fine-tuned outputs. Higher values indicate greater semantic overlap with human references.

RMSE) but yields negligible change in mean error (MAE). The difference in confidence metrics is similarly modest, reflecting the inherent difficulty of this task. Beyond numerical metrics, the semantic similarity between model-generated and human-written reviews was evaluated using BERTScore F1.

The zero-shot model achieved a mean $F1 = 0.838$, while the fine-tuned model reached $F1 = 0.842$, confirming strong overall alignment in meaning and phrasing. Figure 2 illustrates this comparison. Although the improvement is minor, the fine-tuned model exhibits higher structural consistency and greater adherence to the predefined review format (clear separation of “Strengths” and “Weaknesses”), indicating that LoRA fine-tuning primarily reinforces format control and stylistic coherence rather than generating new content knowledge.

4 Final Discussion

The results highlight that LoRA fine-tuning primarily enhanced the structural and stylistic reliability of peer review generation rather than substantially improving numerical prediction accuracy. The fine-tuned model produced more coherent and consistently formatted reviews, achieving a stable BERTScore ($F1 = 0.84$) while slightly reducing variance in rating predictions. However, low R^2 values and weak correlations in confidence estimation confirm that reviewer certainty cannot be inferred solely from textual evidence. These outcomes reflect the subjectivity and sparsity of the dataset and underscore the limits of text-only modeling. Future research should thus expand the dataset with larger and more diverse conference reviews, integrate reviewer metadata and contextual cues, and experiment with multi-task or reinforcement learning frameworks to better capture the underlying rationale behind scoring. Despite these constraints, the study demonstrates that parameter-efficient fine-tuning can yield semantically consistent and well-structured outputs on modest hardware, marking a meaningful step toward scalable, interpretable systems for automated scholarly review assistance.

6 References

- Aczel, B., Barwich, A., Diekman, A., Fishbach, A., Goldstone, R., Gomez, P., Gundersen, O.E. and von Hippel, P., Holcombe, A., Lewandowsky, S., Nozari, N., Pestilli, F., and Ioannidis, J. (2025). The present and future of peer review: Ideas, interventions, and evidence. *Proc. Natl. Acad. Sci. U.S.A.*, 122.
- Bornmann, L. (2011). Scientific peer review. *Annual Review of Information Science and Technology*, 45.
- Gilardi, F., Alizadeh, M., and Kubli, M. (2023). Chatgpt outperforms crowd-workers for text-annotation tasks.
- Hu, E., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., and Chen, W. (2021). Lora: Low-rank adaptation of large language models.
- Tyser, K., Segev, B., Longhitano, G., Zhang, X.-Y., Meeks, Z., Lee, J., Garg, U., Belsten, N., Shporer, A., Udell, M., Te’eni, D., and Drori, I. (2024). Ai-driven review systems: Evaluating llms in scalable and bias-aware academic reviews.
- Zhou, Y., Muresanu, A., Han, Z., Paster, K., Pitlis, S., Chan, H., and Ba, J. (2022). Large language models are human-level prompt engineers.